INCH-POUND

MIL-DTL-32018
13 March 2001
SUPERSEDING
SCD 837AS118, Rev. B
10 August 1990

DETAIL SPECIFICATION

CORD ASSEMBLY, MILD DETONATION, CANOPY PERIPHERY, P/N 837AS118

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the manufacture and acceptance of the MG40 Canopy Periphery Mild Detonation Cord (MDC) Assembly, P/N 83AS118 , referred to in this document as cord assembly.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents specified in Sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplements thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Commander, Indian Head Division, Naval Surface Warfare Center, Engineering Documentation Branch (Code 4230), 101 Strauss Avenue, Indian Head, Maryland 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-P-8184	Plastic Sheet, Acrylic Modified
MIL-DTL-21625	Design and Evaluation of Cartridges for Cartridge Actuated Devices

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-100	Engineering Drawing Practices
MIL-STD-1168	Ammunition Lot Numbering and Ammunition Data Cards

(Unless otherwise indicated, copies of DoD specifications are available from: Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

SPECIFICATIONS

NAVAL SEA SYSTEMS COMMAND (CAGE Code 53711)

WS 5003	Hexanitrostilbene (HNS), Explosive

DRAWINGS

NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003)

837AS118	Cord Assembly, Mild Detonating, Canopy Periphery
837AS829	Cord Assembly, Mild Detonating , Test Fixture

(Application for copies should be addressed to the Commanding Officer, Naval Air Technical Service Facility, 700 Robins Avenue, Philadelphia, PA 19111-5079.)

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2.3 Non-Government publications and drawings. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents, which are DoD adopted, are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)

ASTM E 748	Standard Practice for N-Ray Inspection
ASTM E 1742	Standard Practice for Radiographic Inspection (DoD adopted)

(Application for copies of ASTMs should be sent to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)/ AMERICAN SOCIETY FOR QUALITY CONTROL (ASQC)

ANSI/ASQC Z1.4	Sampling Procedures and Tables for Inspection by Attributes (DoD adopted)

(DoD activities may obtain copies from: Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094. The private sector and other Government agencies may purchase copies from the American Society for Quality Control, P. O. Box 3005, 611 E Wisconsin Ave., Milwaukee, WI 53201-4606.)

OEA AEROSPACE INC. (CAGE CODE 17610)

51852-1	Detonating Cord Assembly, Canopy Periphery

(Application for copies should be addressed to OEA AEROSPACE INC., Highway 12 and Explosive Technology Road, Fairfield, CA 94533.)

MC CORMICK SELPH, INC. (CAGE Code 06331)

819782-1	Detonating Cord Assembly, Canopy Periphery

(Application for copies should be addressed to MC CORMICK SELPH, INC., 3601 Union Road, P.O. Box 6, Hollister, CA 95023.)

2.4 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document,

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however, supercedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection as specified in section 4.3 (see 6.4). A first article inspection shall be required if:

a. One or more years has passed since MDC or similar items have been manufactured,
b. there has been a major turnover in personnel,
c. there have been major changes to the production facility or
d. there has been a transfer of production to a new facility or site.

3.2 Design and construction. The MDC shall be manufactured to contractor drawings 819782-1 or 51852-1, as applicable, and shall meet the requirements of this specification with special effort to ensure inherent reliability so that the units will consistently perform as specified.
3.3 Detonatining cord velocity. The detonation velocity of the detonating cord shall be 5,900 meters per second minimum when tested at –65°F in accordance with 4.4.2. The detonation wave shall propagate through the entire length of the test specimen.

3.4 Core load. The range of the explosive cord loading in grains per foot, divided by the minimal core load shall not exceed 24 percent when calculated from the detonating cord sample data obtained from 4.4.3.

3.5 MDC and booster interface.

3.5.1 Visual inspection. All MDC and booster interfaces manufactured under contract shall be visually inspected and screened for defects (see 4.2.1).

3.5.2 Leakage. All MDC and booster interfaces shall not exhibit a leak rate in excess of 1×10^{-5} cubic centimeters per second of air when tested as specified in 4.4.4.

3.5.3 Radiographic inspection. All MDC and booster interfaces shall be free of imperfections when examined radiographically as specified in 4.4.5 and 4.4.6.

3.5.4 Canopy periphery MDC assembly and subscale assembly.

3.5.4.1 Visual inspection. All assemblies and subscale assemblies manufactured under contract shall be visually inspected and screened for defects (see 4.2.1).

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3.5.4.2 Radiographic inspection. All assemblies and subscale assemblies shall be free of imperfections when examined radiographically as specified in 4.4.5.

3.6 Ballistic requirements. Subscale assemblies from each first article lot and production lot samples shall meet the requirements listed below, when subjected to the tests prescribed in 4.5.

3.6.1 Performance. The MDC shall deliver sufficient energy to sever completely the 0.31 ± 0.03 inch thick cast acrylic plate per MIL-P-8184 over the temperature range of −65°F to +200°F.

3.6.1.2 Misfire. There shall be no misfires (see 4.5.2).

4. VERIFICATION

4.1 Classification of inspections.

a. First article inspection (see 4.2)
b. Conformance inspection (see 4.3)
c. Lot acceptance inspection (see 4.4)

4.2 First article inspection. Before entering into quantity production, an acceptable first article sample of 19 subscales shall be prepared using the same procedures and methods proposed for manufacture of the production lot. An acceptable sample is one that has passed all the tests and examinations of 3.3, 3.4 and 3.5.

4.2.1 Test methodology. Eighteen of the 19 cartridges shall be expended in the tests listed herein and one shall be retained for investigative purposes. First article sample environments shall consist of the following conditioning in table I.

Table I. First article tests.

Environment	Quantity
Temperature & humidity Cycling:	6
Salt spray:	6
Conditioning at –65 °F:	3
Conditioning at +200 °F:	3
Investigative sample:	1

Subscales shall be fired as described in 4.5 and shall meet the functional requirements of 3.6.1.

4.3 Conformance inspection.

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4.3.1 Visual inspection. Visually examine all MDC and booster interfaces prior to final assembly, then examine all full-scale and subscale assemblies. Reject those with inadequate or illegible descriptive marking or serial numbers, dents, deep scratches or other defects, which may prevent proper installation or function.

4.3.2 Corrosion. All parts and components shall be visually inspected prior to assembly to assure there is no corrosion on the hardware. Parts will be inspected to assure parts are free from nicks and blemishes.

4.3.3 Moisture content. Prior to the loading of explosive shall be determined to verify conformance to the 0.3 % maximum moisture content requirement of drawing 837AS118.

4.4 Lot acceptance inspection.

4.4.1 Sampling. To meet the requirements of 3.3 and 3.4 a 12-inch length of cord from each end of the spool and a 12-inch length every 100 feet of the planned cord production shall be removed from the spool for testing. The twelve-inch sample shall be divided into a 9-inch length for velocity tests and the 3-inch sample for the cord loading tests. Failure of any sample to meet the requirements of 3.3 and 3.4 shall result in the rejection of the sampled cord.

4.4.1.1 Sampling for ballistic tests. Components used in the assembly of the subscale assemblies for ballistic test shall be selected from the production lot in accordance with the requirements of Table I, General Inspection Level II of ASQC Z1.4 with the exception that the lot size is limited to 51 minimum and 91 maximum and the sample size shall be 8 subscales.

4.4.2 MDC velocity. The samples shall be tested for velocity measurements by detonating a 9-inch segment, conditioned at –65 °F for 3 hours minimum, with a No. 6 blasting cap with velocity measured over a fixed interval prior to producing any MDC assemblies. The velocity shall meet the requirements of 3.3.

4.4.3 Core load. The amount of core explosive for the 3-inch sample shall be determined to verify conformance with the requirements of 3.4.

4.4.4 Leakage. Each length of MDC and booster interfaces shall be leaked tested in a dry gas leak tester to determine conformance with 3.5.2. Units, which exhibit a leak rate in excess of 1×10^{-5} cubic centimeters per second of air at a pressure differential of 1.0 atmospheres, shall be rejected.

4.4.5 Radiographic inspection. Each length of MDC and booster interface, prior to assembling in the canopy periphery, and each subscale, shall be subjected to radiographic inspection in accordance with ASTM E 1742. The samples shall be arrayed on trays or boards in consecutive numerical order and each radiograph shall carry a permanent identification of the MDC and

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subassembly displayed thereon. The MDC and subscales shall be positioned to give the most revealing cross-section of each unit. All units shall be identified with serial numbers prior to radiographic inspection. The radiographic identification shall include the complete lot number and the span of serial numbers displayed. Discontinuities in serial numbers shall be identified on the plate. Defective MDC and subscales shall be clearly identified on the plate and shall be removed. Radiographs or copies of the entire production lot shall accompany the ballistic sample to the activity conducting the production conformance test (see 6.2).

4.4.5.1 Marriage log. Loaded booster sleeves shall be radiographed (N-Ray) separately and serialed. A marriage log shall be maintained that records each booster with the serial number of the MDC assembly for future reference.

4.4.6 N-Ray inspection. Each MDC and booster interface shall be N-Rayed, in accordance with ASTM E 748, along the full length of the longitudinal axis to verify the presence of explosive, the absence of foreign hydrogenous material, and verify that gaps and compaction densities are within control standards. Any observable imperfection shall be cause for rejection of the unit.

4.4.7 X-Ray inspection. Each MDC length and booster interface and subscale shall be X-Rayed, in accordance with ASTM E 1742, to verify the presence, condition and proper installation of parts. Any observable imperfection shall be cause for rejection of the unit.

4.5 Ballistic tests. Subscale assembly test samples shall be selected in accordance with 4.4.1.1. The samples shall be divided equally between the –65 °F and the +200 °F conditioning groups. The units shall be conditioned for a time period of not less than 4 hours or more than 24 hours. Subscales shall be fired as shown on drawing 837AS829. Units shall be fired within 5 minutes of removal from conditioning. If not fired within 5 minutes, recondition for 2 hours and retest.

4.5.1 Acceptance criteria. The lot shall be acceptable with respect to functional testing if the low and high temperature groups meet the requirements of 3.6.1. Accept on no defects, reject on one or more.

4.5.2 Misfire. A subscale that does not fire under the conditions specified in 4.5 shall be considered unacceptable, unless the misfire is clearly attributable to faulty test equipment or procedure.

4.5.3 Retest. There shall be no retest.

4.5.4 Test failure. If a test failure is attributed to an assignable cause, excluding the test sample, the original test results shall be discarded and that test reconducted.

5. PACKAGING

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5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order, (see 6.2). When actual packaging of material is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.) 1

6.1 Intended use. The MDC covered by this specification is for use in the emergency egress from the AV-8B aircraft. The MG40, Cord Assembly, Mild Detonating, Canopy Periphery, MDC, assembly is designed to completely sever the aircraft canopy 360 degrees along the edge of the canopy in preparation for emergency egress from the aircraft. This explosive, in conjunction with the overhead MDC, fragmentizes the canopy allowing easier seat egress. This MDC is military unique and there is no equivalent commercial application for this design.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).

c. Whether first article inspection is required and, if so, specify the test activity (see 3.1 and 4.3).

d. Size (see 4.4.1) and test activity.

e. Items of data required for each production lot (see 4.4.5).

f. Packaging requirements (see 5.1 and 6.6).

g. Safety requirements (see 6.5).

6.3 Flight critical part. The MDC is a flight critical and the technical data package should be identified and procured as such.

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6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested (see 4.3.1). The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles.

6.4.1 Bids. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Ammunition and explosive safety.

6.5.1 Contractor's responsibility. The safety requirements of the "Contractor's Safety Manual for Ammunition, Explosives and Related Dangerous Material" (DOD 4145.26M) are applicable when this specification is used for acquisition.

6.5.2 Government activity's responsibility. The safety precaution requirements of "Ammunition and Explosives Ashore", OP5 are applicable when this specification is used as part of a description of work to be accomplished by a government activity.

6.6 Contract packaging, packing and marking. The following requirements should be specified in the contract.

6.6.1 Packaging and packing. In addition to the following, Level A and Level C packaging and packing should be in accordance with 49 CFR 100-199. For explanation of terms , see 6.7.

6.6.1.1 Level A. Level A packaging and packing should be used for packaging and packing of all production lots for service use.

6.6.1.1.1 Inner container. A maximum of one assembly should be tightly packaged in KIMPAK dunnage, wrapped in a velostat bag, evacuated of air and heat-sealed. The assembly will then be placed in a cardboard container with enough dry dunnage to provide a tight pack.

6.6.1.1.2 Outer container. One inner container should be packed in the outer container conforming to PPP-B-621, Class 2, Style 4 or 4½ or a box conforming to PPP-B-601, Overseas Type, Style A or B.

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6.6.1.1.3 Level C. Minimum packaging and packing is required for shipping subscales, but a tight pack is required.

6.6.2 Marking.

6.6.2.1 Special marking. Marking of outer containers should be in accordance with CFR 49 171-199.

6.6.2.2 Normal marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages and shipping containers should be marked in accordance with MIL-STD-129.

Inner container marking :--

NSN: 1377-01-136-7571 DODIC: MG40

Nomenclature: Cord Assembly, Mild Detonating, Canopy Periphery

Part Number: 30003-875AS118

Lot Number: (In accordance with MIL-STD-1168)

ICC Explosive Class: C

Outer container marking :--

NSN: 1377-00-136-7571 DODIC: MG40

Nomenclature: Cord Assembly, Mild Detonating, Canopy Periphery

Part Number: 30003-875AS118

Quantity: 1

Gross weight and cube:

Lot number: (In accordance with MIL-STD-1168)

Contract or purchase order number:

"WARNING: EXPLOSIVE POWER DEVICE, CLASS C EXPLOSIVE, HANDLE WITH CARE, KEEP FIRE AWAY"

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The above information should be applied in accordance with the provisions of MIL-STD-129.

6.7 Definitions.

6.7.1 Level A packaging. The degree of preservation and packaging that will afford adequate protection against corrosion, deterioration and physical damage during shipment, handling, indeterminate storage and world wide redistribution.

6.7.2 Level C packaging. The degree of preservation and packaging that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity for immediate use. This level may conform to contractors commercial practice when such meets the requirements of this level.

6.7.3 Level A packing. The degree of packing that will afford adequate protection during shipment, handling, indeterminate storage and worldwide distribution.

6.7.4 Level C packing. The degree of packing that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity for immediate use. This level may conform to contractors commercial practice when such meets the requirements of this level.

6.8 Explanation of terms. For the purposes of this specification, NAVSUPINST 4030.28A should be used to clarify terms.

6.9 Subject term (key word) listing.

Aircraft canopy
Emergency egress
Mild detonating cord
Shape charge

Preparing activity:
Navy – OS
(Project No. 1377-0115)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-DTL-32018	2. DOCUMENT DATE (YYMMDD) 010313

3. DOCUMENT TITLE
CORD ASSEMBLY, MILD DETONATION, CANOPY PERIPHERY, P/N 837AS118

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
C. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) DSN (*If applicable*)	7. DATE SUBMITTED (*YYMMDD*)

8. PREPARING ACTIVITY

a. NAME Commander, Indian Head Division Naval Surface Warfare Center	b. TELEPHONE (*Include Area Code*) (1) Commercial 301-744-4700 (2) DSN 354-4700
c. ADDRESS (*Include Zip Code*) Engineering Documentation Branch (Code 4230) 101 Strauss Avenue Indian Head, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, VA 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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